INDIAN INSTITUTE OF TECHNOLOGY DELHI

DEPTT./CENTRE PHYSICS

CERTIFICATE PAGE

REPORT OF THE PROJECT SHORTLISTED UNDER SURA-2024

Project titled DESIGN AND FABRICATION OF DIFFRACTIVE OPTICAL ELEMENTS (FOR UNCONVENTIONAL IMAGING SYSTEMS)

Submitted by:

Name of the Student	Deptt. /Centre	Entry No.	Contact No. & Email Address (Mobile no.)	Signature of the Students
1. Anushka Pandey	Engineering Physics	2022PH11848	+91 8887820831 ph1221848@iitd.ac.in	
2. Shruti Aparna	Engineering Physics	2022PH11850	+91 8210879969 ph1221850@iitd.ac.in	

Name and Signatures of the Facilitators:

Prof. Kedar Khare ______________________ **Prof. Samaresh Das** _______________________

His/her Deptt./Centre : Optics and Photonics **His/her Deptt./Centre :** Electrical Engineering

Mobile No. of the Facilitator : 0112659 1362

Date of Submission of Report in IRD:

Contents

1. Introduction

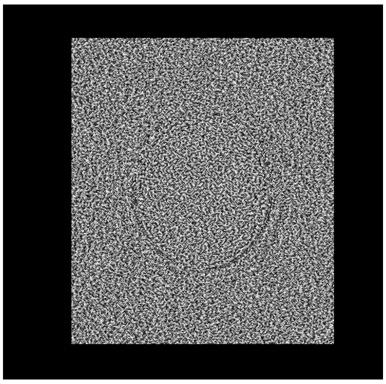


Fig. 1: A phase mask profile

Diffractive optical elements (DOEs) essentially function due to the interplay of the phenomena of diffraction and interference that often display vivid colors. Diffraction is a phenomenon that is observable due to the wave nature of light. The effects of diffraction are more noticeable when light interacts with an interface or an optical element whose features are close to those of the incident wavelength. Diffraction theory allows one to calculate how wavefronts change and how they travel after interaction with such an element.

By manipulating the diffraction patterns through the precise design and fabrication of DOEs, we aim to explore this novel approach in optical engineering. This innovation holds the potential for unconventional imaging systems, allowing for enhanced spatial representation, extended field of view, and depth of focus. The DOEs can be added to existing imaging systems like a microscope or a digital camera as phase masks to enhance imaging capabilities.

The project aims to design and implement custom DOEs tailored to specific applications like displays and projection systems to control light distribution, which can also be used as a Fourier plane filter. These custom-designed DOEs offer the flexibility to shape and manipulate light in ways that conventional optics cannot, providing superior control over the phase and intensity of light in various optical setups. These can then be adapted for multiple functions, such as beam shapers, beam splitters, pattern generators, kinoforms, and gratings. We aim to enhance the performance of these elements, ensuring precise control over light propagation by using advanced optimization algorithms. The ultimate goal is to develop DOEs that contribute to high-resolution imaging, low-latency displays, and enhanced visual experiences, further pushing the boundaries of optical system design in areas like biomedical imaging, augmented reality, and communication technologies.

2. Theory

2.1 The importance of the Fourier phase

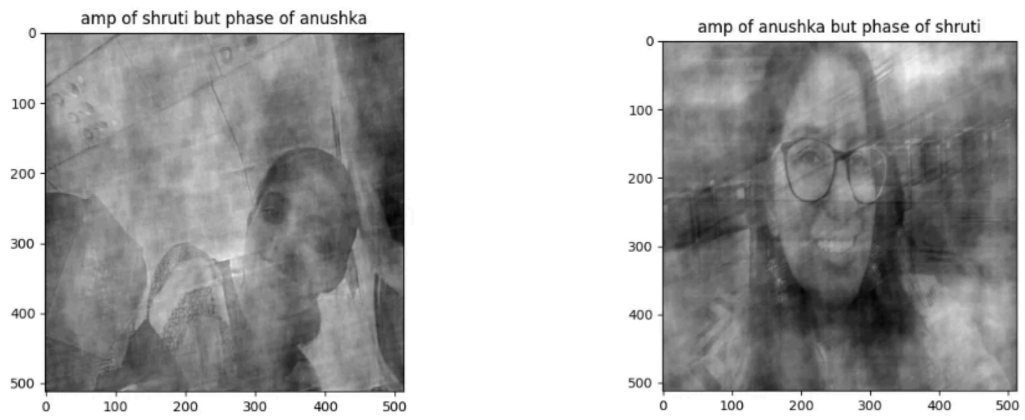


Fig. 2: Two different face-images are Fourier transformed. After swapping their phases, they are inverse Fourier transformed. The result clearly demonstrates the importance of phase information for image recovery.

2.2 Gerchberg-Saxton Algorithm:

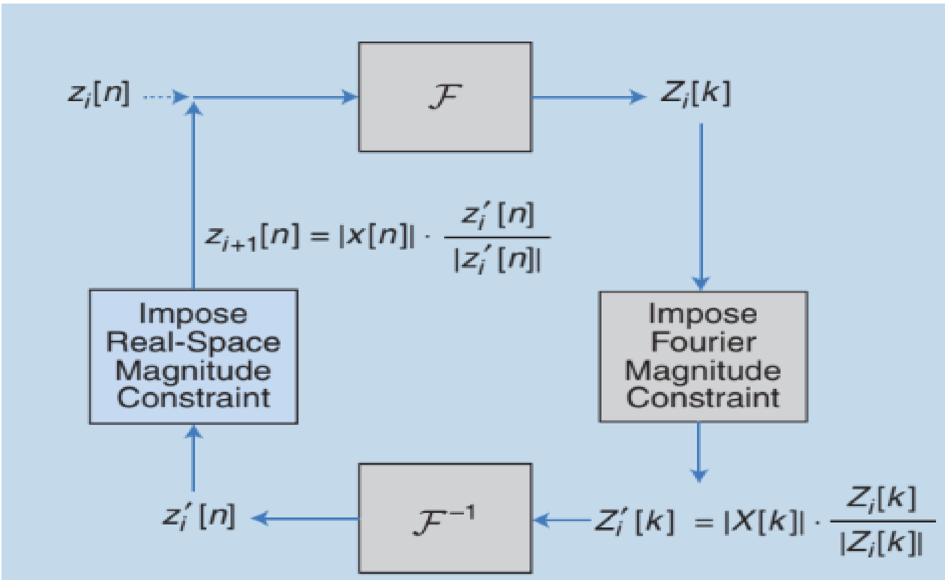


Fig. 3: The block diagrams of the GS algorithm

The GS algorithm helps retrieve the phase of a wavefront by alternating between two known constraints: one in the spatial (image) domain and the other in the Fourier (far-field) domain.

Starting with an initial guess for the phase in the input domain, the algorithm applies a Fourier transform to move to the output domain, where the amplitude is replaced by the desired one while keeping the phase. An inverse Fourier transform is then applied to return to the input domain, where the known amplitude is imposed again. This process is repeated, alternating between the two domains until the phase converges and meets the desired constraints in both domains

Limitations:

1. Slow Convergence: While the algorithm generally converges, the rate of convergence can be slow, especially for complex phase profiles.
2. Local Minima: The algorithm may get stuck in local minima, meaning it may not always find the globally optimal phase distribution, particularly for non-trivial patterns.
3. No Unique Solution: There may not be a unique phase solution for a given intensity distribution, so the algorithm may return different solutions depending on the initialization.

2.3 Optimization of Gerchberg-Saxton Algorithm Using Adam

To enhance the performance and convergence of the Gerchberg-Saxton algorithm, we incorporated Adam optimization, a gradient-based optimization technique that improves convergence speed and accuracy. While the traditional GS algorithm can be slow and prone to getting stuck in local minima, Adam optimizes the phase updates by adjusting the learning rate dynamically for each parameter, based on the first and second moments of the gradients. This approach accelerates convergence and leads to more accurate phase retrieval, especially for complex intensity profiles, by reducing artifacts like speckle noise and improving the quality of the resulting diffractive optical elements (DOEs). By combining Adam with the GS algorithm, we achieved more efficient and reliable solutions for phase mask design.

How Adam Works:

Adam uses two key concepts:

1. First Moment (Mean of Gradients): This is an estimate of the mean of the gradient of the loss function with respect to the parameters, i.e., the average of past gradients.
2. Second Moment (uncentered variance of gradients): This is an estimate of the uncentered variance of the gradient, i.e., the average of the squared gradients.

Adam keeps track of both of these moments during the optimization process and uses them to adaptively adjust the learning rate for each parameter.

3. Materials and experimental methods

3.1 Materials & facilities used:

nominal optics such as lenses, lasers, CCD cameras, Spatial Light modulators, and UV Lithography (NRF, IIT Delhi and CARE, IIT Delhi)

Softwares & python libraries: GDOESII, Ledit, imageio, PIL, matplotlib, numpy, autograd, scipy

3.2 Procedure:

1. We conducted literature surveys on various theories related to diffractive optical elements and how this phenomenon could be manipulated to achieve unconventional results. This was followed by a study of existing algorithms to design DOEs, including the Gerchberg-Saxton algorithm, and methods to optimize the design by error minimization.

2. The GS algorithm was then implemented in Python to calculate the desired phase profile for the DOE. This involved iteratively updating the phase distribution to achieve a targeted intensity pattern at the output plane, based on Fourier domain constraints. To further improve the DOE's performance, we incorporated Adam optimization, reducing speckle noise and refining the convergence of the phase reconstruction.

3. Before fabrication, we tested the design using the spatial light modulator (SLM) in the lab. The SLM allowed us to simulate the DOE's phase profile in real-time, enabling us to verify the diffraction pattern before proceeding with the physical fabrication process.

4. For fabrication, we used the facility of UV lithography available at CARE, IIT Delhi, to transfer the phase profile on a transparent substrate, quartz. Initially, we fabricated a spiral of dots as a test sample, with the entire pattern size measuring 2560 µm x 2560 µm. The smallest feature size achieved within this pattern was 5 µm.

5. After fabrication, we placed the DOE in an optical system to observe the results. The diffracted light created a pattern of points on the image plane according to the DOE's design.

Target Image before optimization Image after optimization

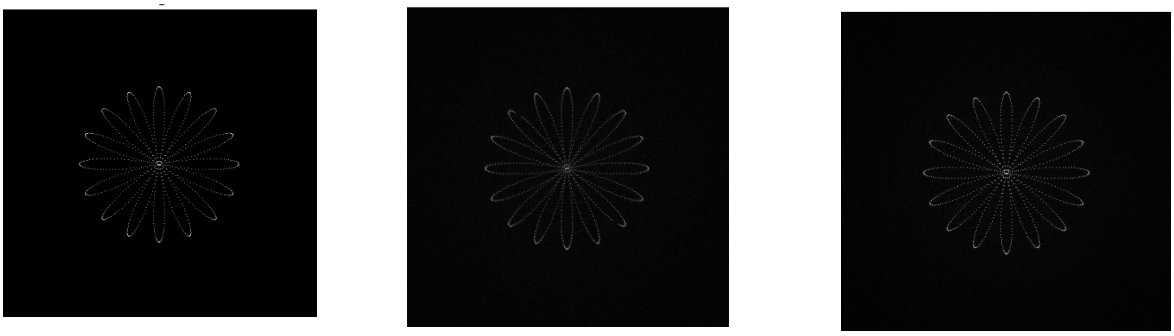


Fig. 4. (left) input image, (middle) output image by applying just GS algorithm, (right) output image after adam optimization

Figure 4 illustrates the output of the Gerchberg-Saxton (GS) algorithm with and without Adam optimization. The input image (left) represents the target pattern. The middle image shows the output produced solely by the GS algorithm, where the reconstructed pattern is generated but suffers from significant noise and artifacts. In contrast, the image on the right demonstrates the result after applying Adam optimization, which considerably improves the output by reducing noise and achieving a closer approximation to the target pattern. The Adam optimization enhances the overall quality, as evidenced by the sharper edges and more defined structure of the final output.

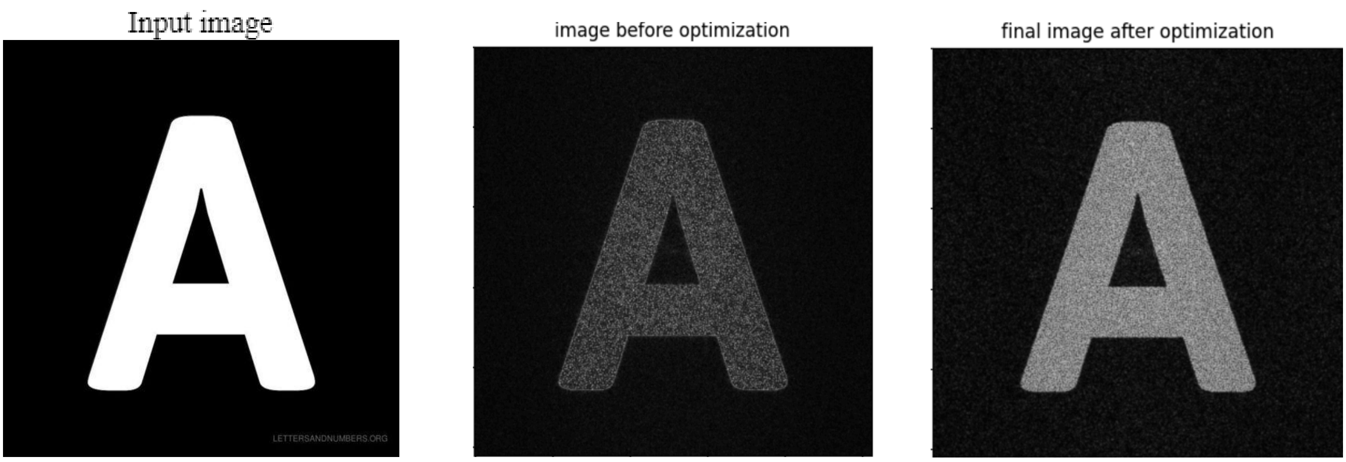


Fig. 5. (left) input image, (middle) output image by applying just GS algorithm, (right) output image after adam optimization

Figure 5 displays the input image (left), the output before optimization (middle), and the final output after applying Adam optimization (right). While the image after optimization closely resembles the target image, with significantly reduced noise, there is still some noticeable speckle present. This becomes more pronounced when dealing with complex images like the one shown. The optimization process enhances the overall similarity between the target and the output, yet the presence of speckle noise suggests that further refinement or additional noise-reduction techniques may be needed, especially for intricate patterns.

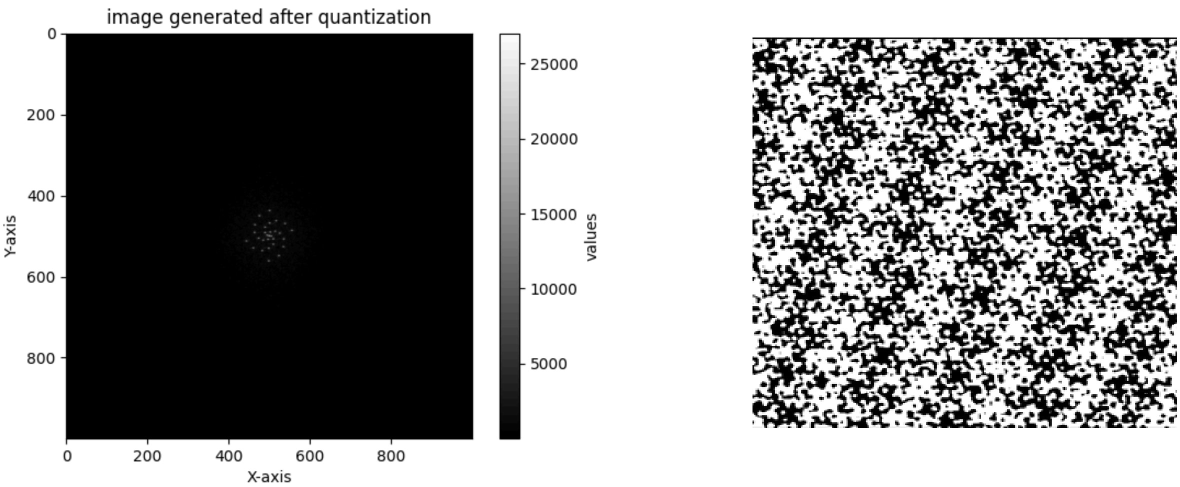


Fig. 6. Two-level quantized image and phase profile

Figure 6 illustrates the output generated after quantization. The need for quantization arises due to the limited precision of spatial light modulators (SLMs), which can typically only display a finite number of discrete phase levels. Without quantization, the continuous range of values produced by the optimization algorithms cannot be accurately represented on the SLM. By applying quantization, we ensure that the resulting phase profile can be properly displayed.

To quantify the improvement after optimization in complex images, we have calculated the following image quality metrics:

1. **Mean Squared Error (MSE)**

This measures the average squared difference between the original and optimized images. A lower MSE value indicates a better optimization result.

Before Optimization: 188,705

After Optimization: 17,026

Interpretation: A **dramatic reduction** in MSE suggests that the optimized image is much closer to the target image in terms of pixel values. This is a positive indication of the success of our optimization process.

2. **Structural Similarity Index (SSIM)**

SSIM assesses perceptual similarity between the original and optimized images, considering luminance, contrast, and structure. A value close to 1 indicates better similarity

Before Optimization: 0.018

After Optimization: 0.038

Interpretation: The **SSIM** has improved, indicating that the structural and perceptual similarity between the optimized image and the target image is better. This improvement suggests that the optimized image has more similar textures, edges, and features to the target image, which often correlates well with visual quality.

3. **Peak Signal-to-Noise Ratio (PSNR)**

This metric provides the ratio between the maximum pixel value and the noise. A higher PSNR value indicates better image quality after optimization.

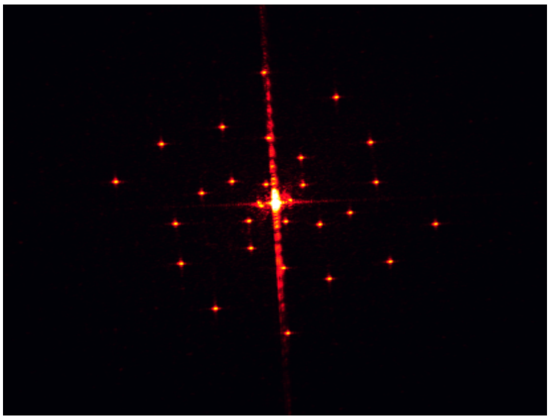
Before Optimization: 17.74 dB

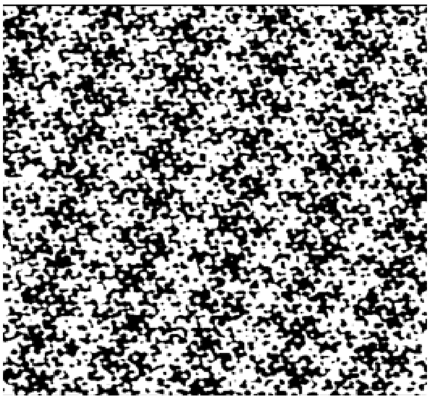
After Optimization: 17.43 dB

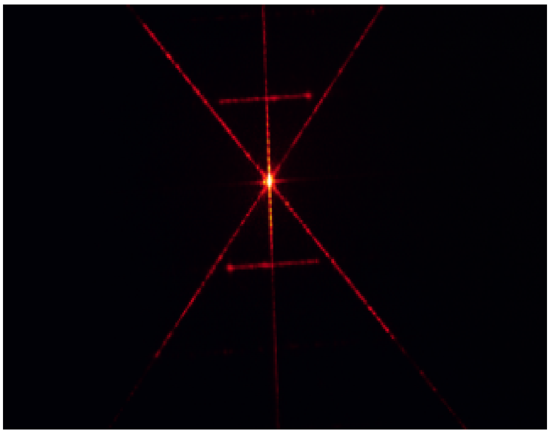
Interpretation: The PSNR values are relatively close. While there is a slight decrease after optimization, it isn't too alarming since PSNR can sometimes behave differently depending on image content. The more notable improvements in other metrics, such as MSE and SSIM, indicate overall enhancement.

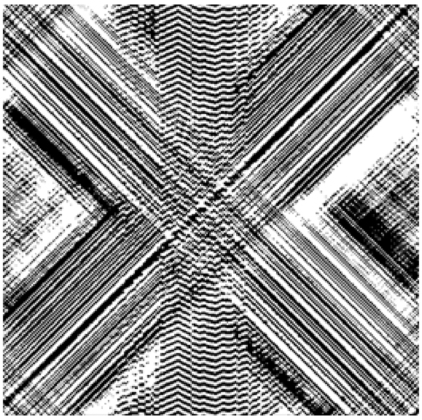
3.4. SLM Results:

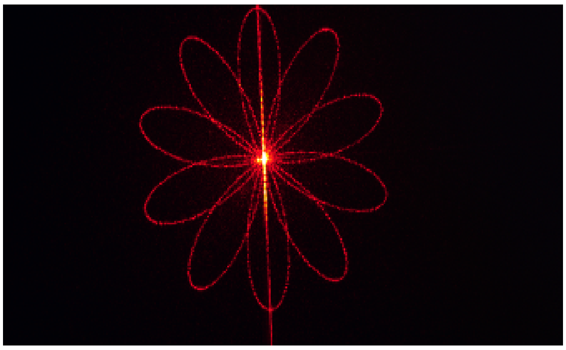
To test the phase profiles on SLM, we first converted the final phase profile matrix to an 8-bit grayscale image:

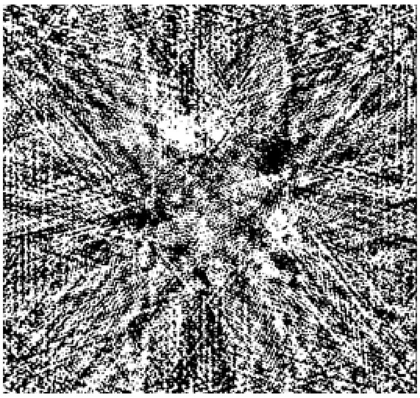


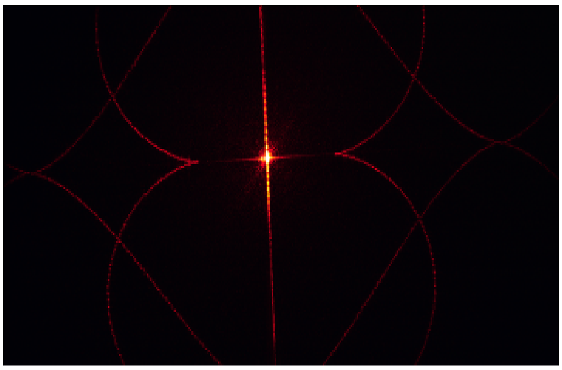












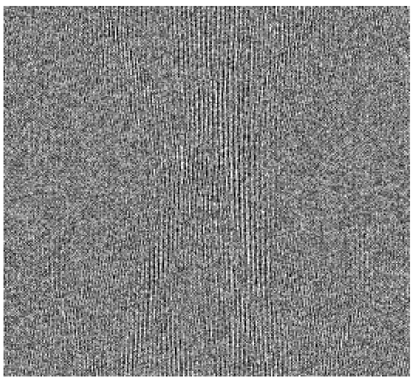


Fig. 7. (left) image on output screen, (right) their respective phase masks

3.5 UV Lithography

For lithography, we quantized the phase profile into two levels and created a GDS file using Ledit. We sent the phase profile for a spiral of dots for fabrication. Before that, to understand the lithography process, we started with the fabrication of rectangular slit patterns.

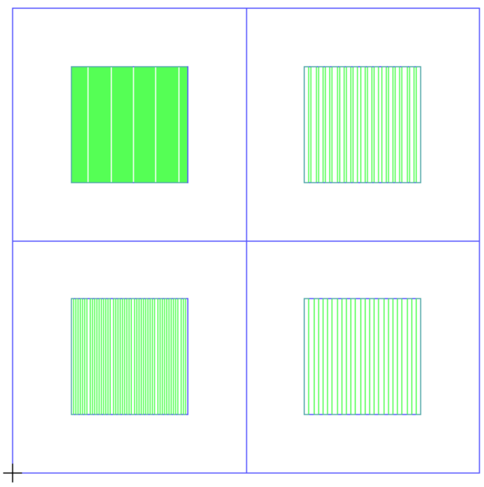


Fig. 8. Design of rectangular slit patterns.

We then tested this phase mask on the optical setup that we designed as shown below:

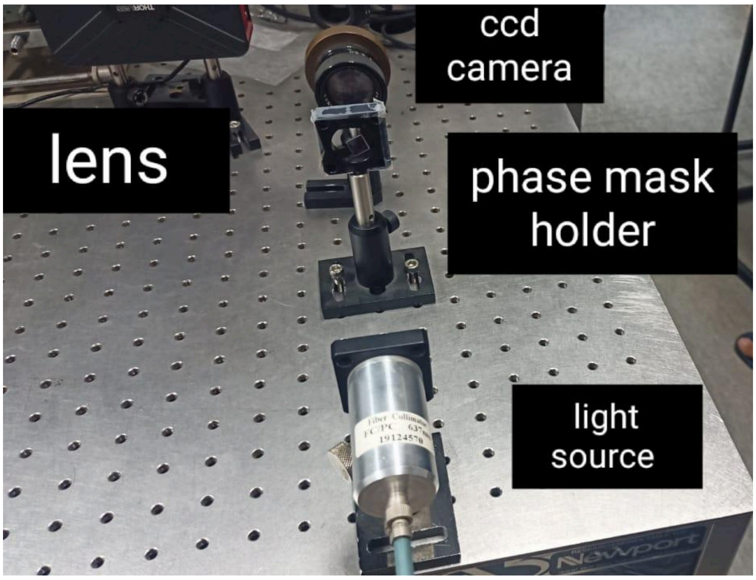


Fig. 9. Testing setup

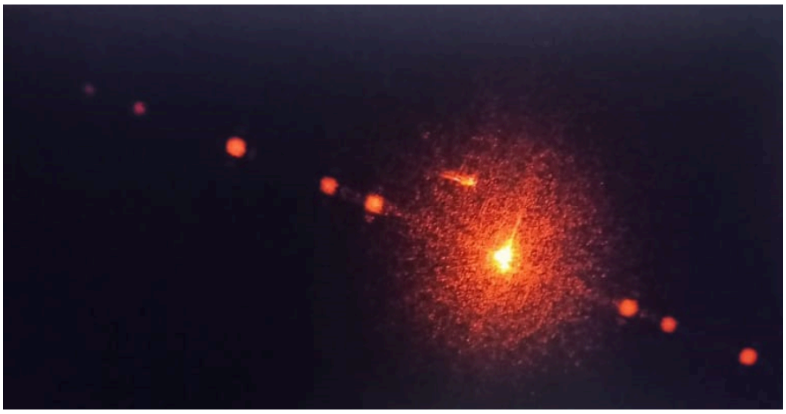


Fig. 10. Output for slits

We then moved to the fabrication of phase masks for spiral dots:

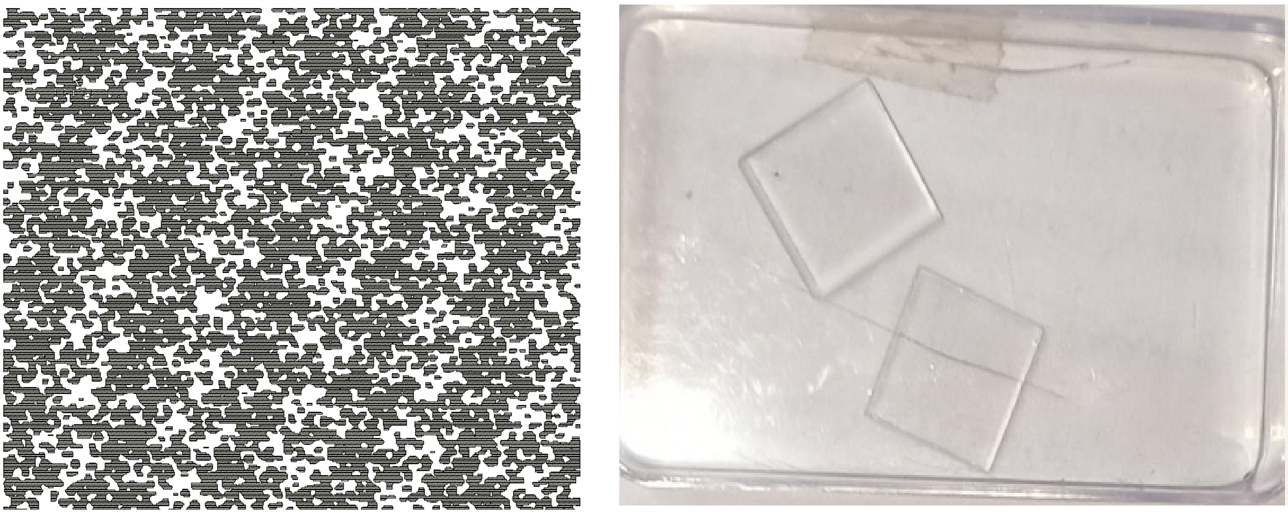


Fig. 11: (left) phase mask for spiral dots designed on ledit, (right) fabricated quartz with the said design

4. Results

1. Developed an advanced iterative algorithm from existing IFTA algorithms for designing phase profiles of a multi-level DOE.
2. Employed optimizing techniques, including gradient descent & Adam optimizer, to refine & get the optimal solution.
3. Converted the final phase profile matrix to an 8-bit grayscale image for testing on the Spatial Light Modulator (SLM).
4. Quantized the phase profile into two levels and created a GDS file using Ledit for precise UV lithography fabrication.

5. Discussion

This project aimed to design and fabricate diffractive optical elements (DOEs) optimized for unconventional imaging systems. The use of iterative Fourier transform algorithms (IFTA) like the Gerchberg-Saxton algorithm, supplemented by optimization techniques such as Adam, played a crucial role in achieving our objectives. The results show significant improvements in phase reconstruction, with a reduction in Mean Squared Error (MSE) from 188,705 to 17,026, confirming the effectiveness of our approach. However, while the improvements in Structural Similarity Index (SSIM) were more modest, the enhancement in image structure and feature preservation is notable.

5.1 Challenges in Fabrication

One of the primary challenges encountered during this project was the fabrication of DOEs using UV lithography. The fabrication process, particularly in achieving the desired phase mask features with sub-micron accuracy, was complex. Another limitation was the two-level quantization of the phase profile, which restricted the fidelity of the reconstructed wavefront.

In addition, minor discrepancies between the simulated and fabricated DOE performance were observed, likely due to imperfections in the lithography process. Variations in the thickness of the photoresist layer and the exposure time might have contributed to these deviations.

5.2 Observations with SLM

The testing of phase profiles on the Spatial Light Modulator (SLM) proved to be a valuable intermediate step. The SLM allowed us to quickly evaluate different phase designs and make necessary adjustments before physical fabrication. We observed that the Adam-optimized phase profiles exhibited less speckle noise and improved convergence compared to those generated solely using the Gerchberg-Saxton algorithm. The overall pattern produced was closer to the target design, which indicates the robustness of the optimization techniques employed.

5.3 Future Improvements

To further enhance the precision and applicability of the fabricated DOEs, a few improvements could be implemented:

1. **Multi-level Quantization:** Incorporating multi-level phase quantization in the fabrication process would allow for finer control over the phase shifts and improve the accuracy of the diffraction patterns.
2. **Advanced Fabrication Techniques:** The use of electron-beam lithography instead of UV lithography could help achieve higher resolution in DOE fabrication, allowing for more intricate patterns and smaller feature sizes.
3. **Exploring Machine Learning Algorithms:** In future iterations of this project, the use of machine learning-based optimization algorithms could be explored to potentially improve the convergence speed and accuracy of the phase retrieval process.
4. **Wider Application Testing:** The DOEs developed in this project could be tested in a variety of real-world applications, such as biomedical imaging and augmented reality systems, to assess their performance in practical scenarios and gather additional data for refinement.

6. Conclusion

In conclusion, this project successfully developed diffractive optical elements (DOEs) optimized for unconventional imaging systems using advanced iterative algorithms and optimization techniques, including Adam. A key metric, the Signal-to-Noise Ratio (SNR), was used to assess image quality post-optimization. While there was a slight decrease in SNR after optimization, the other metrics, such as structural similarity, showed significant improvements, indicating an overall enhancement in image quality. Challenges in fabrication and phase quantization limitations were acknowledged, but the project made considerable progress. Future efforts could focus on multi-level phase quantization, advanced fabrication techniques, and machine learning to further refine DOE performance and expand their applicability in real-world systems.

7. References

- J R Feinup, "Iterative method applied to image reconstruction and computer-generated holograms", Optical Engineering 19(3), 297-305

  https://labsites.rochester.edu/fienup/wp-content/uploads/2019/07/OEngr1980_ITAimRecCGH.pdf

- Anand Vijayakumar, Shanti Bhattacharya, "Design and fabrication of Diffractive optical elements DOEs using MATLAB", SPIE.DigitalLibrary (2017)

- Kedar Khare, "Fourier Optics and Computational Imaging", Wiley (2015)

- E. R. Dowski and W. T. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859–1866 (1995).

  https://opg.optica.org/ao/fulltext.cfm?uri=ao-34-11-1859&id=45359

- Kedar Khare and Ritika Malik, "Single-shot extended field of view imaging using point spread function engineering", J. Opt. Soc. Am. A 40, 1066-1075 (2023)

  https://opg.optica.org/josaa/fulltext.cfm?uri=josaa-40-6-1066&id=530503

- Raghu Dharmavarapu, Shanti Bhattacharya, Saulius Juodkazis, "GDOESII: Software for design of diffractive optical elements and phase mask conversion to GDSII lithography files," *SoftwareX* 9, 126-131 (2019)

  https://www.sciencedirect.com/science/article/pii/S2352711018302711

8. Appendix

- All the codes and results : https://github.com/Khuspd121/SURA-project